

Claude Finance Terminal MASTER PROMPT

Here's the exact prompt I used. You'll need three things:

1. **Claude Code** installed (free at claude.com/claude-code)
2. **Your own Anthropic API key** (console.anthropic.com – for the AI analysis features)
3. **Your own FRED API key** (fred.stlouisfed.org/docs/api – free, for the macro page)

Open Claude Code in an empty folder and paste the prompt below. It'll scaffold the whole app. About 30-45 minutes of back-and-forth gets you a working dashboard. I'm not your advisor and this is for educational use only.

Part 2 – The master prompt (copy this verbatim)

None

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Build me a personal stock market dashboard called "Stock Market Analyst"
using Streamlit. It is for educational and personal-research use only and
will not give buy/sell recommendations to viewers.
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```
# STACK
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- Python 3.9+
- Streamlit for the UI
- yfinance for prices, fundamentals, ETF metadata, news (no API key needed)
- FRED (fredapi) for macro indicators (user supplies their own free key)
- Anthropic Claude API for AI analysis (user supplies their own key)
- plotly for charts, pandas for data

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# SETUP RULES
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- Read keys from a .env file in the project root using python-dotenv. Variables: ANTHROPIC_API_KEY, FRED_API_KEY.
- Add .env, __pycache__, .venv, venv, .DS_Store, data/portfolio.json, and .streamlit/secrets.toml to .gitignore.
- Never hard-code keys. Never print keys. If a key is missing, show a friendly message in the relevant page instead of crashing.
- requirements.txt should pin: streamlit, yfinance, plotly, pandas, fredapi, python-dotenv, anthropic, requests.

FILE STRUCTURE

```
app.py                                -> landing page with quick market
snapshot
pages/1_📈_Market_Pulse.py
pages/2_🔍_Stock_Analyzer.py
pages/3_📊_ETF_Analyzer.py
pages/4_🌐_Macro.py
pages/5_💼_Portfolio.py
pages/6_📰_News.py
lib/market_data.py                   -> yfinance wrappers: get_quote,
get_history, get_quotes_bulk, get_history_bulk, get_etf_details,
get_stock_fundamentals, is_etf, INDEX_TICKERS, SECTOR_ETFS, PERIOD_MAP
lib/charts.py                         -> render_price_chart with 4 views
(Performance, Price, Candlestick, Area), shared split_traces helper
lib/logos.py                         -> ticker -> domain map for top US
companies + base64 data-URL loader from assets/logos/
lib/news.py                          -> ticker_news + market_news using
yfinance + Yahoo RSS fallback
lib/macro.py, lib/rates.py           -> FRED helpers
lib/signals.py                       -> compute technical and fundamental
scores (NOT exposed as buy/sell)
lib/risk.py                          -> ETF risk score (0-100, volatility +
drawdown + concentration based)
lib/etf_peers.py                     -> curated peer groups for cost
comparison
lib/portfolio.py                     -> load/save user holdings to
data/portfolio.json
lib/claude_analyst.py                 -> Anthropic API calls for bull/bear
case, deep analysis, macro pulse-check
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lib/config.py          -> DISCLOSURE string used on every
page footer
assets/logos/          -> downloaded logos for top tickers
scripts/fetch_logos.py -> downloader that tries simple-icons
SVG -> apple-touch-icon -> faviconV2 -> DuckDuckGo, picks best
# CHART RULES (critical for accurate visuals)
- Use REAL INDICES, not ETFs: ^GSPC (S&P 500), ^NDX (Nasdaq 100), ^DJI
(Dow), ^RUT (Russell 2000), ^VIX, ^TNX (10Y yield), GC=F (gold futures),
CL=F (crude WTI), BTC-USD, DX-Y.NYB (DXY).
- Performance view splits the line at the zero baseline: green segments
above, red below, with linearly interpolated zero crossings. Same logic
for Area view, baselined at starting price.
- For 1D intraday charts, the baseline must be YESTERDAY'S CLOSE, not
today's first 5-min bar. Pass it as an optional baseline_price parameter
to render_price_chart. Otherwise overnight gaps make the chart look up
when the market is actually down.
- Dark mode throughout (plotly_dark template).
- Show end-of-period return as a colored badge anchored at the last point.
# COMPLIANCE FRAMING (NON-NEGOTIABLE)
- NO buy/hold/sell signals or recommendations anywhere in the UI.
- The Stock Analyzer has a "Snapshot" section with two gauges labeled
"Technical strength" and "Fundamental quality" (0-100 each) and a
descriptive "At a glance" panel showing factual buckets: Trend
(above/below 200DMA), Momentum (RSI bucket), 52-week range position,
Profitability (ROE tier), Leverage (D/E tier), Volatility (beta vs
market), Valuation (P/E tier).
- Use neutral words: "Firm momentum," "Higher than market," "Low
multiple." Avoid "Buy," "Sell," "Recommend," "Should."
- Every page ends with this disclosure: "This dashboard is for educational
and informational purposes only. It is not financial advice, not a
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recommendation to buy or sell any security, and is not personalized to your situation. Consult a licensed advisor before making investment decisions."

MARKET PULSE PAGE

- Period selector (1D, 5D, 1M, 3M, 6M, YTD, 1Y, 3Y, 5Y, 10Y, 20Y, 30Y, Max).
- Grid of cards for the 10 index/asset tickers above, each with a tiny sparkline. Sparkline splits green/red at the period starting price; 1D sparklines prepend yesterday's close as the baseline bar.
- Big S&P 500 (^GSPC) chart with view toggle.
- Sector heatmap bar chart for the 11 SPDR sector ETFs (XLK, XLF, XLV, XLE, XLI, XLY, XLP, XLU, XLRE, XLB, XLC).
- Top gainers / losers / most active in three columns, each row showing company logo + ticker + name + price + % change colored green/red.
- Top 3 market headlines from the last 24 hours, single-column compact layout with title, publisher, time-ago, 2-line summary.

STOCK ANALYZER PAGE

- Ticker input + period selector.
- Header row with 64px company logo, name, ticker, sector/industry, and metric tiles for Price, Market Cap, Trailing P/E, Beta.
- Big chart with the 4 view toggles and volume overlay.
- Snapshot section: descriptive caption on top, then three columns. Left = "At a glance" card with the 7 neutral-language chips. Middle = "Technical strength" gauge (0-100 with red-to-green band, needle, score number below, brief subtitle "Trend, momentum, position vs averages"). Right = "Fundamental quality" gauge same style with subtitle "Margins, returns, leverage, growth".
- Driver bullet lists below each gauge (factual observations only: "Price above 200DMA", "ROE 30%", etc).

- Key statistics grid: Valuation, Profitability, Balance sheet, Trading, Income, Analyst.

- Business summary in an expander.

- AI analysis tabs: Bull/Bear case (Claude), Deep analysis (Claude), Recent headlines.

ETF ANALYZER PAGE

- Same chart pattern with prev-close baseline for 1D.

- Returns table (YTD, 3Y avg, 5Y avg, 3Y beta).

- Risk gauge (0-100, color bands from conservative to very aggressive, needle style).

- Sector breakdown horizontal bar chart.

- Top holdings as a two-column list with company logo + symbol + name + weight (NOT a plain dataframe).

- Peer comparison table sorted by expense ratio, plus a "cheaper alternatives" callout that computes savings in basis points and dollars on a \$100K position.

MACRO PAGE

- FRED indicators: GDP, unemployment, CPI YoY, core CPI, Fed funds, 10Y-2Y spread, retail sales, industrial production.

- Yield curve chart.

- Claude-generated macro pulse-check (button-triggered).

PORTFOLIO PAGE

- User enters tickers + shares + cost basis.

- Save to data/portfolio.json (gitignored).

- Show current value, total return, allocation pie, sector breakdown.

- Risk score for the portfolio.

- Claude deep analysis button.

NEWS PAGE

- Tab 1: market headlines (aggregated). Tab 2: by-ticker search.

LOGOS

- Write scripts/fetch_logos.py that downloads logos for the top ~125 US tickers and ETFs into assets/logos/<TICKER>.<ext>. Try sources in this order: simple-icons via jsdelivr CDN (SVG, post-process to fill="#ffffff" for dark mode), vectorlogo.zone, apple-touch-icon from the company's domain, Google's faviconV2 at size=256, DuckDuckGo icons. Pick the best per ticker.

- At render time, encode the local file as a base64 data URL (no live third-party fetch).

UX DETAILS

- Sidebar shows " Market Analyst" branding on every page.
- Larger base font for accessibility (font-size: 17px on main content).
- Use st.metric, st.segmented_control, and st.columns for layouts.
- Cache yfinance calls with @st.cache_data and reasonable TTLs (60s for quotes, 300s for history).

START HERE

Begin with app.py, lib/config.py (DISCLOSURE constant), lib/market_data.py (INDEX_TICKERS, SECTOR ETFs, PERIOD_MAP, get_quote, get_history, get_quotes_bulk), and lib/charts.py (render_price_chart with the green/red split and the baseline_price parameter). Then build Market Pulse. Then build Stock Analyzer with the Snapshot panel. Then the rest.

Confirm setup with me before installing dependencies. Always show me the file paths you're creating or editing.